

APB AES Core: RTL-to-GDSII Physical Sign-Off Report

Design: apb_aes | Technology Node: OSU 350nm CMOS (osu035) | Sign-Off Status: PASS

1. Executive Sign-Off Summary

Metric	Result	Status
Design Rule Check (DRC)	0 Violations	PASS
Layout vs. Schematic (LVS)	Circuits match uniquely	PASS
Max Operational Frequency	381.25 MHz ((T_{clk} = 2.62))	PASS
Setup Time / Slack	Met ((T_{setup} = 0.199))	PASS
Tapeout Mask Export	layout/apb_aes.gds (5.1 MB)	VERIFIED

2. Toolchain Pipeline

- **Logic Synthesis:** Yosys (osu035 cell mapping)
- **Placement & Routing:** GrayWolf & Qrouter (Metal 1–3)
- **DRC & GDSII Generation:** Magic VLSI
- **Static Timing Analysis (STA):** Vesta (Post-routing extraction)
- **LVS Equivalence Verification:** Netgen (Topological match)

3. Physical Verification (LVS Proof)

Final result: Circuits match uniquely.
Cell pin lists are equivalent.
Device classes apb_aes and apb_aes are equivalent.

4. Static Timing Analysis Sign-Off

- **Critical Path:** _3957_/CLK -> _3839_/D
- **Data Path Delay:** 2.373 ns
- **Clock Skew:** 0.049 ns
- **Maximum Frequency:** 381.25 MHz

5. Tapeout Deliverables Inventory

- `apb_aes.gds` (5.1 MB) - Fabricated mask database
- `apb_aes.def` (1.3 MB) - DEF standard layout
- `lvs_comp.out` (24 KB) - Netgen LVS equivalence proof
- `timing_sta.log` (42 KB) - Post-route timing report
- `apb_aes_tapeout_signoff.tar.gz` (1.3 MB) - Sign-off compressed archive